Electricity and magnetism

Mark scheme and worked solutions for 120 questions: 60 multiple choice, 60 structured.

1 · Part 1

Core · Simple phenomena of magnetism · [1]

Answer: B

Worked solution

Like poles repel and unlike poles attract. The back of X is an N pole, so the front of Y needs an S pole to be attracted.

2 · Part 1

Core · Simple phenomena of magnetism · [1]

Answer: D

Worked solution

An unmagnetised piece of iron is attracted by either pole of a magnet, and so is the opposite pole of another magnet. Attraction alone therefore cannot tell you whether the bar is a magnet. Repulsion is the only sure test: it needs like poles, and only a magnet has poles of its own.

3 · Part 1

Core · Simple phenomena of magnetism · [1]

Answer: A

Worked solution

Remember the pair: soft iron = temporary (magnetises and demagnetises easily), steel = permanent (harder to magnetise but keeps its magnetism).

4 · Part 1

Core · Simple phenomena of magnetism · [1]

Answer: C

Worked solution

Magnetic materials are attracted by a magnet and can be magnetised: iron, steel, nickel and cobalt. Most other metals, including copper, aluminium and brass, are non-magnetic.

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5 · Part 1 Core Simple phenomena of magnetism [1]

Answer: A

Worked solution

The direction of a magnetic field at a point is the direction of the force on an N pole there. The compass N pole is repelled by the magnet's N pole, so it is pushed further east and the needle points east.

6 · Part 1 Core Simple phenomena of magnetism [1]

Answer: C

Worked solution

An electromagnet is magnetic only while a current flows, so it can be turned on to pick the scrap up and off to drop it. Permanent magnets cannot be switched off.

7 · Part 1 Extended Simple phenomena of magnetism [1]

Answer: B

Worked solution

The relative strength of a magnetic field is shown by how close together the field lines are. Lines are closer at P, so the field is stronger at P.

8 · Part 1 Core Simple phenomena of magnetism [1]

Answer: D

Worked solution

When a magnetic material is near a magnet it becomes an induced magnet. The end nearest the magnet's N pole becomes an S pole, so the far end is an N pole. This N pole attracts the second nail, which is also induced, so a chain of nails can hang from the magnet.

9 · Part 1 Core Electric charge [1]

Answer: A

Worked solution

Start with what you know: Z is positive. Unlike charges attract, so Y is negative. Like charges repel, so X is negative too.

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10 · Part 1 Core Electric charge [1]

Answer: D

Worked solution

Only negative charges (electrons) are transferred when solids are rubbed together. A negative rod has more electrons than before, so electrons must have moved from the cloth onto the rod. The cloth is left positively charged.

11 · Part 1 Core Electric charge [1]

Answer: C

Worked solution

In a metal, some electrons are not attached to any one atom. These free electrons can drift through the metal when a voltage is applied, which is what makes copper a good conductor. In an insulator, the electrons are held to their atoms and cannot move through the material.

12 · Part 1 Extended Electric charge [1]

Answer: B

Worked solution

Each kind of field is named after what feels the force in it: charges in an electric field, magnetic poles in a magnetic field, masses in a gravitational field.

13 · Part 1 Extended Electric charge [1]

Answer: B

Worked solution

Physicists chose a positive "test" charge to define electric field direction. A negative charge, such as an electron, feels a force in the opposite direction to the field.

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14 · Part 1 Extended Electric charge [1]

Answer: D

Worked solution

Outside a charged conducting sphere the field lines are straight and radial, like spokes of a wheel. Their direction is the way a positive charge would be pushed: away from a positive sphere, towards a negative one.

15 · Part 1 Extended Electric charge [1]

Answer: C

Worked solution

Step 1: field direction is the direction of the force on a positive charge. A positive charge at Y would be repelled, so the field points away from the point charge.

Step 2: the particle is negative, so its force is opposite to the field, towards the point charge. This agrees with "unlike charges attract".

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16 · Part 1 Core Simple phenomena of magnetism [4]

Mark scheme

(a) **[1]**

- a magnetic material is attracted by a magnet / can be magnetised, but a non-magnetic material is not attracted / cannot be magnetised (1)

ALLOW magnetic materials are affected by magnets; non-magnetic materials are not

(b) **[3]**

- aluminium: no attraction or repulsion at either end / no effect (1)
- iron: both ends are attracted (1)
- steel magnet: one end is attracted and the other end is repelled (1)

ALLOW steel magnet: repelled at its N pole end and attracted at its S pole end

REJECT for iron, "repelled"

Worked solution

(a) A magnetic material (such as iron, steel, nickel or cobalt) is attracted by a magnet and can be magnetised. A non-magnetic material (such as aluminium) is not attracted and cannot be magnetised.

(b) The aluminium rod is non-magnetic, so nothing happens at either end. The unmagnetised iron rod has magnetism induced in it, so whichever end is tested is attracted. The steel magnet has its own poles: its S pole end is attracted to the N pole, and its N pole end is repelled. Testing both ends is what tells the iron and the magnet apart, because only a magnet can show repulsion.

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17 · Part 1 Core Simple phenomena of magnetism [6]

Mark scheme

(a) **[2]**

- magnetism is induced in the rods / the rods become (induced) magnets because they are touching / near the magnet (1)
- (the bottom end of each rod becomes a pole and) this attracts the tacks / magnetism is induced in the tacks, which are attracted (1)

(b) **[1]**

- N pole (1)

(c) **[3]**

- (most of) the tacks fall off rod X (1)
- (most of) the tacks stay on rod Y (1)
- soft iron loses its (induced) magnetism easily / is a temporary magnet, but steel keeps its magnetism / becomes a permanent magnet (1)

ALLOW a few tacks may stay on X / a few may fall from Y

Worked solution

(a) A magnetic material close to a magnet becomes magnetised: this is induced magnetism. Each rod becomes a magnet while it touches the strong magnet, so its bottom end attracts the iron tacks (which are magnetised by induction too).

(b) Unlike poles attract, so the end touching the S pole becomes an N pole. The bottom end is therefore an S pole.

(c) Soft iron makes a temporary magnet: when the strong magnet is removed, rod X loses almost all its magnetism and the tacks drop off. Steel makes a permanent magnet: rod Y keeps much of its magnetism, so the tacks stay attached.

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18 · Part 1 Core Simple phenomena of magnetism [6]

Mark scheme

(a) **[1]**

- a region in which a magnetic pole experiences a force (1)

(b) **[4]**

- place the magnet on the paper and draw round it / mark its position (1)
- place the compass near the N pole and mark a dot at the end of the needle (the end the N pole of the needle points to) (1)
- move the compass so that the other end of the needle is over the dot, mark a new dot, and repeat many times (1)
- join the dots with a smooth line and add an arrow in the direction the N pole of the compass needle points (away from N, towards S) (1)

(c) **[1]**

- the direction of the field (1)

Worked solution

(a) A magnetic field is the region around a magnet where another magnetic pole would feel a force.

(b) The compass needle lines up with the field. Mark where the N end of the needle points, move the compass along so its tail sits on that mark, and keep going. The chain of dots traces a field line from the N pole round to the S pole. The direction of the field is the direction of the force on an N pole, which is the way the compass N pole points, so the arrow goes from the N pole towards the S pole.

(c) Iron filings line up along the field lines and show the pattern, but each filing has no labelled N end, so they do not show which way the field points.

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19 · Part 1 Core Simple phenomena of magnetism [6]

Mark scheme

(a) **[1]**

- aluminium is non-magnetic / is not attracted by a magnet (1)

(b) **[1]**

- it can be switched off to drop the cans / the magnetism can be turned on and off (1)

ALLOW its strength can be changed

(c) **[2]**

- any two from: compass needle; loudspeaker; microphone; refrigerator door seal; magnetic catches / door catches; electric motor; generator
- first correct use (1)
- second correct use (1)

REJECT crane / relay / electric bell / circuit breaker (uses of electromagnets)

(d) **[2]**

- soft iron loses its magnetism (easily) when the current is switched off (1)
- so the cans are released / drop off; a steel core would stay magnetised and the cans would stay attached (1)

ALLOW soft iron is a temporary magnet; steel is a permanent magnet

Worked solution

(a) Only magnetic materials are attracted by a magnet. Steel is magnetic; aluminium is not, so the aluminium cans are not lifted.

(b) The crane has to let go of the steel cans over the bin. An electromagnet loses its magnetism when the current is switched off; a permanent magnet cannot be switched off.

(c) Permanent magnets are used wherever a magnet must always be on, for example in compasses, loudspeakers and refrigerator door seals.

(d) The core must stop being magnetic as soon as the current stops. Soft iron is a temporary magnet and loses its magnetism easily, so the cans drop. Steel would keep its magnetism and the cans would stay stuck to the crane.

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20 · Part 1 Extended Simple phenomena of magnetism [5]

Mark scheme

(a) **[2]**

- each magnet has its own magnetic field / the field of each magnet acts on the other magnet (1)
- the two fields interact (combine), and this interaction produces the (attractive) force on each magnet (1)

ALLOW field lines from the N pole of P pass into the S pole of Q, joining the two fields

(b) **[1]**

- the field is stronger between the magnets than beyond the ends / closer lines mean a stronger field (1)

(c) **[2]**

- the field lines in the gap are further apart / the field in the gap is weaker (1)
- the force between the magnets is smaller, because the interacting fields are weaker (at the greater distance) (1)

REJECT force becomes repulsive

Worked solution

(a) A magnet does not need to touch another to push or pull it. Each magnet is surrounded by a magnetic field. Where the fields of P and Q overlap they interact, and this interaction is what produces the force on each magnet. Here the field from P's N pole runs into Q's S pole, and the magnets attract.

(b) The spacing of field lines shows relative field strength: closely spaced lines mean a strong field, widely spaced lines a weak one. So the field is strongest in the gap between the magnets.

(c) With a bigger gap, each magnet's field is weaker where the other magnet is, so the lines in the gap are more widely spaced. The interaction between the fields is weaker, so the attractive force is smaller. The poles facing each other are unchanged, so the force is still attractive.

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21 · Part 1 Core Simple phenomena of magnetism [6]

Mark scheme

(a) **[1]**

- N pole (1)

(b) **[2]**

- like poles (N and N) face each other, so the rings repel / there is an upward magnetic force on the upper ring (1)
- the upward magnetic force is equal to the weight of the upper ring / the forces on it are balanced (1)

(c) **[2]**

- the ring moves back up (and may bounce up and down before settling) (1)
- the repulsive force is stronger when the magnets are closer, so it is now greater than the weight (1)

ALLOW forces no longer balanced; upward force bigger

(d) **[1]**

- the upper ring is attracted and moves down to the lower ring / it rests on (sticks to) the lower ring (1)

ALLOW because unlike poles now face each other

Worked solution

(a) The ring floats, so it must be pushed up. Repulsion needs like poles, so the bottom face of the upper ring is an N pole, facing the N pole on top of the lower ring.

(b) The repulsion gives an upward force on the upper ring. Its weight acts downwards. When the ring is at rest these two forces are equal and opposite.

(c) Pushing the ring closer makes the magnetic force stronger, so it is bigger than the weight and the ring is pushed back up towards where the forces balance.

(d) Turning the lower ring over puts its S pole on top. Now an S pole faces an N pole, unlike poles attract, and the upper ring is pulled down onto the lower ring.

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22 · Part 1 Core Simple phenomena of magnetism [4]

Mark scheme

(a) **[1]**

- iron / soft iron / steel (1)

ALLOW nickel; cobalt; any named magnetic metal

REJECT aluminium; copper; brass

(b) **[2]**

- magnetism is induced in the plate (by the magnet) / the plate becomes an induced magnet (1)
- the end of the plate nearest the magnet becomes the opposite pole to the pole facing it, and unlike poles attract (1)

(c) **[1]**

- steel keeps its magnetism / makes a permanent magnet, whereas soft iron would lose its magnetism (1)

Worked solution

(a) The plate only has to be a magnetic material, so iron or steel is suitable.

(b) When a magnetic material is near a magnet, magnetism is induced in it. If the magnet's N pole faces the plate, the near side of the plate becomes an S pole. Unlike poles attract, so the plate is pulled onto the magnet.

(c) The catch must work for years, so the magnet must be permanent. Steel keeps its magnetism; soft iron loses it easily and is only used for temporary magnets.

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23 · Part 1 Core Electric charge [5]

Mark scheme

(a) **[1]**

- positive and negative (1)

(b) **[2]**

- electrons (move) (1)
- from the jumper / wool to the balloon (1)

REJECT positive charges / protons move

(c) **[2]**

- positive (1)
- the jumper has lost electrons / now has fewer electrons than protons (1)

Worked solution

(a) There are two kinds of charge: positive and negative.

(b) When solids are rubbed together, only electrons are transferred. The balloon becomes negative, so it has gained electrons; they came from the wool.

(c) The jumper started with equal amounts of positive and negative charge. It has lost electrons, so it is left with more positive charge than negative charge: it is positively charged.

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24 · Part 1 Core Electric charge [5]

Mark scheme

(a) **[3]**

- rub the ruler (firmly) with the dry woollen cloth (1)
- hold the rubbed ruler close to (above) the small pieces of tissue paper (1)
- the pieces of paper are attracted / jump up to the ruler, showing it is charged (1)

ALLOW compare with an unrubbed ruler, which does not attract the paper

(b) **[2]**

- hang one charged ruler from the thread (tied at its middle) so that it can turn freely (1)
- bring the other charged ruler close to one end; the hanging ruler moves away / is repelled, so the charges are the same (1)

Worked solution

(a) Rubbing with the cloth charges the ruler by friction. A charged object attracts small light uncharged objects, so the tissue paper jumps up to the ruler. An unrubbed ruler does not do this, which shows the effect comes from the charge.

(b) Like charges repel and unlike charges attract. The rulers are made of the same material and charged in the same way, so they should have the same type of charge. Hanging one ruler from the thread lets it move easily; when the second charged ruler is brought near, the hanging ruler swings away. Repulsion shows the charges are the same type.

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25 · Part 1 Core Electric charge [7]

Mark scheme

(a) **[3]**

- connect the battery, the lamp and the crocodile clips in series using the leads, leaving a gap between the two clips (1)
- attach each sample in turn between the two crocodile clips (to complete the circuit) (1)
- if the lamp lights the sample is a conductor; if the lamp does not light it is an insulator (1)

(b) **[2]**

- conductors: steel nail and copper wire (1)
- insulators: plastic ruler and glass rod (1)

(c) **[2]**

- steel / metals contain free electrons that can move through the material (and carry the current) (1)
- in glass the electrons are held tightly / are not free to move, so there is no current (1)

Worked solution

(a) Make a series circuit with a gap in it: battery, lamp and two crocodile clips. Clipping a sample across the gap completes the circuit. If the sample lets charge flow, there is a current and the lamp lights.

(b) The metals (steel and copper) are conductors, so the lamp lights. Plastic and glass are insulators, so the lamp stays off.

(c) In a metal, some electrons are free to move between the atoms, so charge can flow through it. In an insulator such as glass, all the electrons are bound to their atoms and cannot move through the material.

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26 · Part 1 Core Electric charge [5]

Mark scheme

(a) **[2]**

- plastic is an insulator (1)
- its electrons are not free to move, so the (transferred) charge stays where it is on the rod / cannot flow away (1)

(b) **[2]**

- copper is a conductor / contains free electrons that can move (1)
- the charge flows (through the rod and) through the student's hand / body (to the ground / Earth), because the body also conducts (1)

ALLOW electrons flow onto or off the rod through her hand

(c) **[1]**

- hold the copper rod by an insulating handle / wear an insulating (e.g. rubber) glove (1)

ALLOW mount the rod on an insulating stand

Worked solution

(a) Rubbing transfers electrons between the cloth and the rod. In an insulator like plastic the electrons cannot move through the material, so the extra (or missing) electrons stay on the rubbed area and the rod stays charged.

(b) Copper has free electrons, so charge moves easily through it. The student's body is also a conductor, so the charge flows away through her hand and the rod does not stay charged.

(c) If the copper rod is held with an insulator, the charge has no conducting path to escape through, so the rod can stay charged.

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27 · Part 1 Extended Electric charge [5]

Mark scheme

(a) **[1]**

- coulomb / C (1)

(b)(i) **[1]**

- east (1)

(b)(ii) **[1]**

- west (1)

(c) **[2]**

- electric field direction: the direction of the force on a positive charge (at that point) (1)
- magnetic field direction: the direction of the force on an N pole (of a magnet at that point) (1)

ALLOW similarity: both are defined by the direction of the force on a chosen test object

Worked solution

(a) Electric charge is measured in coulombs (C).

(b) The field direction is, by definition, the direction of the force on a positive charge, so a positive charge is pushed east. An electron is negative, so its force is in the opposite direction: west.

(c) Both definitions use the direction of a force on a test object. For an electric field the test object is a positive charge; for a magnetic field it is the N pole of a magnet.

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28 · Part 1 Extended Electric charge [6]

Mark scheme

(a) **[2]**

- uniform field / parallel, equally spaced field lines (perpendicular to the plates) (1)
- directed from the positive (upper) plate to the negative (lower) plate / vertically downwards (1)

(b) **[2]**

- upwards (1)
- the force on a negative charge is opposite to the field direction / the negative drop is attracted to the positive plate (and repelled by the negative plate) (1)

(c) **[2]**

- no change (in size or direction) (1)
- the field between the plates is uniform / the same everywhere between the plates (1)

Worked solution

(a) Between two oppositely charged parallel plates, away from the edges, the field lines are straight, parallel and evenly spaced: the field is uniform. Field direction is the direction of the force on a positive charge, which would be pushed away from the positive upper plate towards the negative lower plate, so the field points downwards.

(b) The drop is negative, so its force is opposite to the field: upwards, towards the positive plate.

(c) A uniform field has the same strength and direction at every point between the plates. So wherever the drop is (away from the edges), the electric force on it is the same.

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29 · Part 1 Extended Electric charge [4]

Mark scheme

(a) **[2]**

- radial / straight lines spreading out from the sphere (as if from its centre) (1)
- pointing outwards, away from the sphere (1)

(b) **[2]**

- towards the sphere (1)
- the field points away from the sphere; the force on a negative charge is opposite to the field direction (1)

ALLOW unlike charges attract, for the second mark only if field direction is also referred to

Worked solution

(a) Outside a charged conducting sphere the field lines are radial, like the spokes of a wheel. A positive charge would be repelled by the positive sphere, so the lines point away from it.

(b) A negative charge feels a force opposite to the field direction. The field points away from the sphere, so the force on the dust particle is towards the sphere. This agrees with the rule that unlike charges attract.

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30 · Part 1 Core Electric charge [7]

Mark scheme

(a) **[1]**

- positive (1)

(b) **[2]**

- electrons are transferred (by friction) (1)
- from the hair to the comb (1)

> **REJECT** positive charges move from comb to hair

(c) **[2]**

- every strand has the same (positive) charge (1)
- like charges repel (so the strands push each other apart) (1)

(d) **[2]**

- zero / no overall charge (1)
- charge is only transferred (electrons move from hair to comb); the negative charge gained by the comb equals the positive charge left on the hair (1)

Worked solution

(a) The comb gained negative charge, so the hair must have lost it and is positively charged.

(b) When two solids rub together only electrons move. Electrons moved from the hair onto the comb, making the comb negative and leaving the hair positive.

(c) All the strands lost electrons, so they all carry positive charge. Like charges repel, so each strand pushes the others away and the hair stands apart.

(d) No charge is made or destroyed during rubbing; electrons simply move from one object to the other. The comb's extra negative charge exactly matches the hair's positive charge, so together they still have no overall charge.

1 · Part 2 Core Electric current [1]

Answer: B

Explanation

A metal contains free (delocalised) electrons that drift through the wire when there is a p.d. across it, so B is right. The positive ions in a metal are fixed in place (A), protons are inside the nuclei and do not move through the metal (C), and the atoms do not travel along the wire (D).

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2 · Part 2 Core Electric current [1]

Answer: D

Explanation

An alternating current keeps changing direction, so D is right. A describes a steady direct current. B is still direct current, because the direction never reverses even though the size changes. C is wrong because a cell produces direct current.

3 · Part 2 Core Electric current [1]

Answer: A

Explanation

0.3 A is 300 mA, which is beyond the end of the 0 to 100 mA and 0 to 10 mA ranges (C and D), so those would go off the scale. Both 0 to 1 A and 0 to 10 A can measure 0.3 A, but the smaller range, 0 to 1 A, gives a larger deflection or more digits and so a more precise reading, so A is right and B is wrong.

4 · Part 2 Extended Electric current [1]

Answer: C

Worked solution

Convert the time to seconds: 2.0 min = 120 s. Current is charge per unit time: I = Q / t = 36 C / 120 s = 0.30 A.

5 · Part 2 Core Electromotive force and potential difference [1]

Answer: A

Explanation

A is the syllabus definition of e.m.f. B is the definition of potential difference across a component. C describes current (charge per second). D describes power (energy per second).

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6 · Part 2 Core Electromotive force and potential difference [1]

Answer: B

Explanation

A voltmeter measures the p.d. between two points, so it is connected across the lamp, in parallel with it (B). Connecting it in series (A or D) puts it in the path of the current, and connecting it across the ammeter (C) measures the p.d. across the ammeter, not the lamp.

7 · Part 2 Extended Electromotive force and potential difference [1]

Answer: D

Worked solution

p.d. is the work done per unit charge: V = W / Q = 3450 J / 15 C = 230 V.

8 · Part 2 Core Electromotive force and potential difference [1]

Answer: C

Explanation

Both e.m.f. and p.d. are measured in volts, so C is right. The ampere (A and B) is the unit of current. The joule (D) is the unit of work or energy, not of e.m.f.

9 · Part 2 Extended Electromotive force and potential difference [1]

Answer: B

Worked solution

Step 1: charge moved, Q = I × t = 0.20 A × 50 s = 10 C. Step 2: from E = W / Q, W = E × Q = 9.0 V × 10 C = 90 J. Check: W = IEt = 0.20 × 9.0 × 50 = 90 J.

10 · Part 2 Core Resistance [1]

Answer: A

Worked solution

R = V / I = 6.0 V / 0.40 A = 15 Ω.

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11 · Part 2 Core Resistance [1]

Answer: D

Explanation

For wires of the same metal, resistance is greater for a longer wire and greater for a thinner wire (smaller cross-sectional area). The long, thin wire (D) therefore has the greatest resistance. A is short and thick, the lowest resistance. B is thin but short. C is long but thick.

12 · Part 2 Extended Resistance [1]

Answer: C

Worked solution

Halving the length halves the resistance: 12 Ω becomes 6.0 Ω. Three times the area makes the resistance one third: 6.0 Ω / 3 = 2.0 Ω. Check in one step: 12 × (1 / 2) × (1 / 3) = 2.0 Ω.

13 · Part 2 Core Electrical energy and electrical power [1]

Answer: A

Worked solution

P = I × V = 0.60 A × 4.5 V = 2.7 W.

14 · Part 2 Core Electrical energy and electrical power [1]

Answer: B

Explanation

The kilowatt-hour is a unit of energy: the energy transferred by a 1 kW appliance in 1 hour (B). A wrongly calls it a power. C describes only 1 kJ of energy (1 kW for 1 s). D describes a rate of energy transfer, which is a power, not an energy.

15 · Part 2 Core Electrical energy and electrical power [1]

Answer: C

Worked solution

90 minutes = 1.5 h. Energy each day = 2.0 kW × 1.5 h = 3.0 kW h. Energy for 7 days = 3.0 × 7 = 21 kW h. Cost = 21 kW h × 30 cents per kW h = 630 cents.

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16 · Part 2 Core Electric current [4]

Mark scheme

(a) **[1]**

- charge (1)

ALLOW charged particles / electrons (in a metal)

(b) **[2]**

- the metal contains free electrons (1)
- (when there is a p.d. across the wire) the free electrons move / drift through the wire (1)

REJECT movement of protons or positive ions

(c) **[1]**

- the electrons repeatedly change direction / move backwards and forwards (instead of drifting in one direction only) (1)

Worked solution

(a) An electric current is a flow of charge. In a metal wire the charge is carried by electrons.

(b) The atoms in a metal have some electrons that are not tied to any one atom. These free electrons can move through the metal. When the cell is connected, the free electrons drift through the wire, and this flow of charge is the current. The positive ions stay in fixed positions.

(c) With a cell the current is direct current: the electrons drift in one direction only. With alternating current the direction of the current keeps reversing, so the free electrons move one way, then the other way, again and again.

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17 · Part 2 Core Electric current [4]

Mark scheme

(a) **[1]**

- in series (with the motor) (1)

(b) **[2]**

- 0 to 1.0 A (1)
- the largest expected current (0.80 A) is less than the maximum of this range, so the reading stays on the scale, AND the smaller range gives a larger deflection / more precise reading (than 0 to 5.0 A) (1)

(c) **[1]**

- any one from: easier to read / no need to judge the pointer position between scale marks / no parallax error / reading given to more precise value (1)

Worked solution

(a) An ammeter measures the current through a component, so all the current must pass through it. It is connected in series with the motor.

(b) Both ranges can measure up to 0.80 A without going off the scale. The 0 to 1.0 A range is better because the pointer moves much further across the scale for the same current, so the reading can be made more precisely. On the 0 to 5.0 A range the pointer would only move a small way and the reading would be less precise.

(c) A digital meter shows the value as numbers, so there is no need to judge where a pointer lies between marks.

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18 · Part 2 Extended Electric current [6]

Mark scheme

(a) **[1]**

- (the) charge passing a point per unit time (1)

> **ALLOW** rate of flow of charge
>
> **REJECT** flow of charge (only)

(b) **[3]**

- t = 40 × 60 = 2400 s (1)
- Q = It OR Q = 1.5 × 2400 (1)
- 3600 C (1)

> **ECF** from a wrong time conversion for the last two marks

(c)(i) **[1]**

- from the positive terminal to the negative terminal (1)

(c)(ii) **[1]**

- from the negative terminal to the positive terminal (1)

Worked solution

(a) Current is the charge passing a point per unit time, I = Q / t.

(b) First change the time to seconds: 40 min × 60 = 2400 s. Rearrange I = Q / t to give Q = I × t = 1.5 A × 2400 s = 3600 C. Check: 3600 C / 2400 s = 1.5 A.

(c) By convention, current is shown flowing out of the positive terminal, round the circuit, and back into the negative terminal. The free electrons are negatively charged, so they actually flow the opposite way: out of the negative terminal and back into the positive terminal.

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19 · Part 2 Extended Electric current [6]

Mark scheme

(a) **[3]**

- t = 1.2 ms = 0.0012 s OR 1.2×10^{-3} s (1)
- I = Q / t OR I = 0.024 / 0.0012 (1)
- 20 A (1)

ECF from a wrong time conversion for the last two marks

(b) **[2]**

- number = 0.024 / (1.6×10^{-19}) (1)
- 1.5×10^{17} (1)

(c) **[1]**

- 5.0 A (1)

ECF (a) answer divided by 4

Worked solution

(a) Convert the time: 1.2 ms = 1.2×10^{-3} s = 0.0012 s. Current is charge per unit time: I = Q / t = 0.024 C / 0.0012 s = 20 A. Check: 20 A × 0.0012 s = 0.024 C.

(b) Each electron carries 1.6×10^{-19} C. The number of electrons is the total charge divided by the charge on one electron: 0.024 / (1.6×10^{-19}) = 1.5×10^{17} electrons. Check: $1.5 \times 10^{17} \times 1.6 \times 10^{-19}$ = 0.024 C.

(c) The same charge flows in 4 times as long (4.8 ms / 1.2 ms = 4), so the current is 4 times smaller: 20 A / 4 = 5.0 A. Check: 0.024 C / 0.0048 s = 5.0 A.

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20 · Part 2 Core Electromotive force and potential difference [5]

Mark scheme

(a) **[2]**

- work done by a unit charge (1)
- passing through the component / lamp (1)

ALLOW energy transferred (per unit charge) for work done

(b) **[1]**

- volt / V (1)

(c) **[2]**

- the battery does 3.0 J of electrical work (1)
- for each unit charge it moves around the complete circuit (1)

ALLOW per coulomb for per unit charge

Worked solution

(a) The potential difference across the lamp is the work done by a unit charge passing through the lamp. It tells you how much energy each unit of charge transfers in the lamp.

(b) p.d. is measured in volts (V).

(c) e.m.f. is the electrical work done by a source in moving a unit charge around a complete circuit. An e.m.f. of 3.0 V means the battery does 3.0 J of work on each unit charge that it moves around the whole circuit.

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21 · Part 2 Core Electromotive force and potential difference [4]

Mark scheme

(a) **[1]**

- in parallel (with the resistor) / across the resistor (1)

(b) **[2]**

- 0 to 3 V (1)
- 2.4 V is less than 3 V so it stays on the scale, AND the pointer moves further across the scale / gives a more precise reading (1)

(c) **[1]**

- the pointer would go beyond the end of the scale / off the scale (and the voltmeter could be damaged) (1)

ALLOW the voltmeter could not measure 9 V

Worked solution

(a) A voltmeter measures the p.d. between two points, so it is connected across the resistor, in parallel with it.

(b) Starting on the highest range is a safe first step. The reading of 2.4 V is below 3 V, so the 0 to 3 V range can be used. On this range the pointer moves much further across the scale for 2.4 V, so the reading is more precise.

(c) 9 V is far more than the 3 V maximum of that range, so the pointer would try to go past the end of the scale and the meter could be damaged. The 0 to 15 V range should be used.

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22 · Part 2 Extended Electromotive force and potential difference [6]

Mark scheme

(a) **[2]**

- W = VQ OR W = 12 × 1500 (1)
- 18 000 J (1)

ALLOW 1.8×10^4 J

(b) **[3]**

- t = 5.0 × 60 = 300 s (1)
- I = Q / t OR I = 1500 / 300 (1)
- 5.0 A (1)

ECF from a wrong time conversion for the last two marks

(c) **[1]**

- 12 J (1)

Worked solution

(a) p.d. is the work done per unit charge, V = W / Q, so W = V × Q = 12 V × 1500 C = 18 000 J.

(b) Convert the time: 5.0 min = 300 s. I = Q / t = 1500 C / 300 s = 5.0 A. Check: power = IV = 5.0 × 12 = 60 W, and 60 W × 300 s = 18 000 J, which agrees with (a).

(c) A p.d. of 12 V means 12 J of work is done by each coulomb of charge passing through the headlamp (12 V = 12 J / C).

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23 · Part 2 Extended Electromotive force and potential difference [8]

Mark scheme

(a) **[2]**

- Q = W / E OR Q = 120 / 6.0 (1)
- 20 C (1)

(b) **[2]**

- I = Q / t OR I = 20 / 80 (1)
- 0.25 A (1)

ECF from (a)

(c) **[2]**

- W = VQ OR W = 5.5 × 20 (1)
- 110 J (1)

ECF from (a)

(d) **[2]**

- 120 − 110 = 10 J (1)

ECF 120 J minus (c)

- in the connecting leads (they are heated) (1)

ALLOW inside the battery

Worked solution

(a) e.m.f. is the work done by the source per unit charge, E = W / Q, so Q = W / E = 120 J / 6.0 V = 20 C.

(b) I = Q / t = 20 C / 80 s = 0.25 A.

(c) p.d. is the work done per unit charge passing through the resistor: W = V × Q = 5.5 V × 20 C = 110 J. Check: E = IVt = 0.25 × 5.5 × 80 = 110 J.

(d) The battery did 120 J of work, but only 110 J was transferred in the resistor. The other 120 − 110 = 10 J was transferred somewhere else in the circuit. The long connecting leads have some resistance, so they are heated slightly by the current.

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24 · Part 2 Core Electromotive force and potential difference [4]

Mark scheme

(a) **[1]**

- any one from: both are measured in volts / both are work (or energy) per unit charge (1)

> **ALLOW** both are measured with a voltmeter (connected in parallel)

(b) **[1]**

- e.m.f. is the work done by a source in moving unit charge around a complete circuit, but p.d. is the work done by unit charge passing through a component (1)

> **ALLOW** e.m.f. is for a source (energy given to charge), p.d. is for a component (energy transferred from charge)

(c) **[2]**

- current / charge is not used up: the same charge flows back to the cell / the current is the same at all points in the circuit (1)
- it is energy that is transferred from the charge to the motor (and then to the surroundings) (1)

Worked solution

(a) Both e.m.f. and p.d. are measured in volts, and both describe work done per unit charge.

(b) e.m.f. describes a source: it is the work done by the solar cell in moving a unit charge around the whole circuit. p.d. describes a component: it is the work done by a unit charge as it passes through the motor.

(c) Charge flows all the way around the circuit and back to the solar cell, so it is not used up and the current going into the motor is the same as the current coming out. What the motor uses is energy. The charge carries energy from the cell to the motor, and the motor transfers it into kinetic energy and, finally, into the surroundings.

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25 · Part 2 Core Resistance [7]

Mark scheme

(a) **[4]**

- wire in series with the battery, switch and ammeter (1)
- voltmeter in parallel with / across the wire (1)
- record the current (from the ammeter) and the p.d. (from the voltmeter) (1)
- adjust the variable resistor to obtain several pairs of readings AND calculate R = V / I for each / find the average (1)

ALLOW variable resistor in series with the wire

(b) **[2]**

- R = V / I OR R = 4.0 / 0.80 (1)
- 5.0 Ω (1)

(c) **[1]**

- any one from: to calculate an average / to reduce the effect of random errors / to check the readings are consistent / to spot an anomalous reading (1)

Worked solution

(a) Connect the wire, the ammeter, the variable resistor and the switch in series with the battery, so the same current passes through the ammeter and the wire. Connect the voltmeter across the ends of the wire, in parallel with it. Close the switch and record the ammeter reading (current) and the voltmeter reading (p.d.). Change the variable resistor to get several different pairs of readings. Work out R = V / I for each pair and find the average.

(b) R = V / I = 4.0 V / 0.80 A = 5.0 Ω.

(c) A single reading could be affected by a random error. Several readings let her check that they agree, spot any anomalous result and take an average, which gives a more reliable value of resistance.

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26 · Part 2 Core Resistance [5]

Mark scheme

(a) **[2]**

- R = V / I OR R = 9.0 / 0.36 (1)
- 25 Ω (1)

(b)(i) **[1]**

- (the thicker wire has a) smaller resistance (1)

(b)(ii) **[1]**

- (the current is) larger / greater than 0.36 A (1)

(c) **[1]**

- the resistance increases (1)

Worked solution

(a) R = V / I = 9.0 V / 0.36 A = 25 Ω. Check: 25 Ω × 0.36 A = 9.0 V.

(b)(i) A thicker wire has a larger cross-sectional area, and a larger cross-sectional area gives a smaller resistance.

(b)(ii) The battery p.d. is the same but the resistance is smaller, so the current is larger (I = V / R).

(c) A longer wire of the same metal and thickness has a larger resistance.

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27 · Part 2 Extended Resistance [7]

Mark scheme

(a) **[2]**

- resistance proportional to length OR R = 8.0 × 1.5 / 2.0 (1)
- 6.0 Ω (1)

(b) **[3]**

- cross-sectional area is one quarter of the first wire's / area ∝ diameter^2 (1)
- resistance inversely proportional to area OR R = 8.0 × 4 (1)
- 32 Ω (1)

REJECT 16 Ω (area taken as halved) for the last two marks

(c) **[2]**

- I = V / R OR I = 3.0 / 6.0 (1)
- 0.50 A (1)

ECF from (a)

Worked solution

(a) Resistance is directly proportional to length. Resistance per metre = 8.0 Ω / 2.0 m = 4.0 Ω per metre, so 1.5 m has a resistance of 4.0 × 1.5 = 6.0 Ω.

(b) The cross-sectional area of a wire is the area of a circle, $\pi d^2 / 4$, so it depends on the diameter squared. Halving the diameter makes the area $(1 / 2)^2 = 1 / 4$ of the original. Resistance is inversely proportional to area, so the resistance becomes 4 times larger: 8.0 Ω × 4 = 32 Ω. A common mistake is to say the area halves, which gives 16 Ω.

(c) I = V / R = 3.0 V / 6.0 Ω = 0.50 A. Check: 0.50 A × 6.0 Ω = 3.0 V.

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28 · Part 2 Core Electrical energy and electrical power [6]

Mark scheme

(a) **[2]**

- energy is transferred (electrically, by the current) from the supply to the fan motor (1)
- the motor transfers it to kinetic energy (of the blades and air) and thermal energy / sound, which is finally transferred to the surroundings (as thermal energy) (1)

(b) **[2]**

- P = IV OR P = 0.75 × 24 (1)
- 18 W (1)

(c) **[2]**

- E = IVt OR E = 0.75 × 24 × 25 × 60 (1)
- 27 000 J (1)

ECF (b) × 1500 s

ALLOW 27 kJ, 2.7×10^4 J

Worked solution

(a) The current carries energy from the supply to the fan motor. The motor transfers this energy to the kinetic energy of the spinning blades and moving air, with some wasted as thermal energy and sound. In the end, all of the energy spreads out into the surroundings, warming them slightly.

(b) P = I × V = 0.75 A × 24 V = 18 W.

(c) Convert the time: 25 min = 25 × 60 = 1500 s. E = I × V × t = 0.75 A × 24 V × 1500 s = 27 000 J. Check: 18 W × 1500 s = 27 000 J.

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29 · Part 2 Core Electrical energy and electrical power [7]

Mark scheme

(a) **[1]**

- the energy transferred by a 1 kW appliance in 1 hour (1)

ALLOW energy transferred when 1 kW is used for 1 h

(b) **[2]**

- energies 1.5 kW h, 0.60 kW h and 2.4 kW h OR powers converted to kW and multiplied by times (1)
- 4.5 kW h (1)

(c) **[2]**

- 4.5 × 30 × 28 (1)
- 3780 cents (1)

ECF from (b)

ALLOW 3800 cents, $37.80

(d) **[2]**

- 1.5 kW h × 3.6×10^6 OR 3000 W × 0.50 × 3600 s (1)
- 5.4×10^6 J (1)

ALLOW 5 400 000 J

Worked solution

(a) One kilowatt-hour is the energy transferred by a 1 kW appliance used for 1 hour.

(b) Change each power to kilowatts by dividing by 1000, then multiply by the time in hours. Oven: 3.0 kW × 0.50 h = 1.5 kW h. Television: 0.12 kW × 5.0 h = 0.60 kW h. Lights: 0.40 kW × 6.0 h = 2.4 kW h. Total = 1.5 + 0.60 + 2.4 = 4.5 kW h each day.

(c) Energy in 30 days = 4.5 × 30 = 135 kW h. Cost = 135 kW h × 28 cents per kW h = 3780 cents.

(d) Energy = power × time, with power in watts and time in seconds. 0.50 h = 0.50 × 3600 = 1800 s. E = 3000 W × 1800 s = 5 400 000 J = 5.4×10^6 J. Check: 1 kW h = 1000 W × 3600 s = 3.6×10^6 J, and $1.5 \times 3.6 \times 10^6 = 5.4 \times 10^6$ J.

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30 · Part 2 Core Electrical energy and electrical power [8]

Mark scheme

(a) **[2]**

- I = P / V OR I = 2400 / 240 (1)
- 10 A (1)

REJECT 0.010 A (power not converted to watts) for the second mark

(b) **[2]**

- R = V / I OR R = 240 / 10 (1)
- 24 Ω (1)

ECF from (a)

(c) **[2]**

- E = IVt OR E = 10 × 240 × 150 (1)
- 360 000 J (1)

ECF from (a)

ALLOW 3.6×10^5 J, 360 kJ

(d) **[2]**

- I = 120 / 24 = 5.0 A (1)
- P = IV = 5.0 × 120 = 600 W (1)

ECF from (b)

Worked solution

(a) Convert the power: 2.4 kW = 2400 W. From P = IV, I = P / V = 2400 W / 240 V = 10 A.

(b) R = V / I = 240 V / 10 A = 24 Ω.

(c) Convert the time: 2.5 min = 150 s. E = I × V × t = 10 A × 240 V × 150 s = 360 000 J. Check: 2400 W × 150 s = 360 000 J.

(d) The resistance stays 24 Ω, so on 120 V the current is I = V / R = 120 V / 24 Ω = 5.0 A. The power is P = IV = 5.0 A × 120 V = 600 W. Halving the p.d. halves the current as well, so the power is one quarter of 2400 W, which agrees.

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1 · Part 3 Core Series and parallel circuits [1]

Answer: B

Worked solution

There is only one path round a series circuit, so the same charge passes every point each second. The current between the second and third lamps equals the current between the battery and the first lamp: 0.30 A.

2 · Part 3 Core Series and parallel circuits [1]

Answer: C

Worked solution

For cells in series, add the e.m.f.s of cells pointing the same way and subtract any that point the opposite way: 1.5 V + 1.5 V + 1.5 V − 1.5 V = 3.0 V.

3 · Part 3 Core Series and parallel circuits [1]

Answer: A

Worked solution

Combined resistance in series = R1 + R2 + R3 = 12 + 18 + 30 = 60 Ω. The combined resistance of a series arrangement is always larger than any single resistor in it.

4 · Part 3 Core Series and parallel circuits [1]

Answer: A

Worked solution

Adding a second path in parallel always makes it easier for charge to flow, so the combined resistance is smaller than the smaller resistor (10 Ω). Of the options only 8.0 Ω is below 10 Ω. (Extended candidates can check: 1/R = 1/10 + 1/40 = 5/40, so R = 8.0 Ω.)

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5 · Part 3 Core Series and parallel circuits [1]

Answer: D

Worked solution

Each lamp in a parallel lighting circuit is connected directly across the supply, so each gets the full mains p.d. and works at normal brightness, and each branch can be switched independently. If one lamp breaks, the others stay lit.

6 · Part 3 Extended Series and parallel circuits [1]

Answer: B

Worked solution

The sum of the currents entering a junction equals the sum of the currents leaving it. In: 0.50 + 0.20 = 0.70 A. Known out: 0.15 + 0.25 = 0.40 A. Third wire: 0.70 − 0.40 = 0.30 A. Check: 0.15 + 0.25 + 0.30 = 0.70 A.

7 · Part 3 Extended Series and parallel circuits [1]

Answer: C

Worked solution

Method 1: p.d. across each parallel branch = 12 V. I = V / R gives 2.0 A in the 6.0 Ω resistor and 4.0 A in the 3.0 Ω resistor; supply current = 2.0 + 4.0 = 6.0 A.

Method 2: 1/R = 1/6.0 + 1/3.0 = 3/6.0, so R = 2.0 Ω; I = 12 / 2.0 = 6.0 A. Both methods agree.

8 · Part 3 Core Action and use of circuit components [1]

Answer: D

Worked solution

In a series circuit every resistor has the same current. Since V = IR, the resistor with the largest resistance has the largest p.d. across it: the 20 Ω resistor.

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9 · Part 3 Extended Action and use of circuit components [1]

Answer: A

Worked solution

V1 = 9.0 V − 6.0 V = 3.0 V. R2 = R1 × V2 / V1 = 2.0 kΩ × 6.0 / 3.0 = 4.0 kΩ.

Check another way: current = 3.0 V / 2.0 kΩ = 1.5 mA; R2 = 6.0 V / 1.5 mA = 4.0 kΩ.

10 · Part 3 Core Electrical safety [1]

Answer: C

Worked solution

P = IV, so I = P / V = 1610 W / 230 V = 7.0 A. Choose the fuse with the smallest rating that is larger than the normal working current: 10 A. Check: 230 V × 7.0 A = 1610 W.

11 · Part 3 Core Electrical safety [1]

Answer: B

Worked solution

If the switch were in the neutral wire, the appliance would stop working when switched off, but its parts would still be connected to the live wire and could give a shock. A switch in the live wire cuts the appliance off from the high p.d., so it is safe to touch when switched off.

12 · Part 3 Core Electrical safety [1]

Answer: D

Worked solution

Damp conditions are hazardous because water can make a conducting path between live parts and a person. A current through the body can cause a serious or fatal electric shock.

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13 · Part 3 Core Electromagnetic induction [1]

Answer: C

Worked solution

It is the change in the magnetic field linking the coil that matters, not the field itself. In A, B and D the magnet and the coil move relative to each other, so an e.m.f. is induced. In C nothing moves, so the meter reads zero even though the coil is in a strong field.

14 · Part 3 Core Electromagnetic induction [1]

Answer: B

Worked solution

The induced e.m.f. is larger when the magnet moves faster, when the magnet is stronger, and when the coil has more turns. Only B is on this list.

15 · Part 3 Extended Electromagnetic induction [1]

Answer: A

Worked solution

Hold the right hand with the thumb, first finger and second finger at right angles to each other. Point the thumb in the direction of motion (east) and the first finger in the direction of the field (downwards). The second finger points north, so the induced current in the wire flows towards the north.

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16 · Part 3 Core Series and parallel circuits [5]

Mark scheme

(a) **[1]**

- 4.5 V (1)

(b) **[1]**

- 9.0 Ω (1)

(c) **[2]**

- I = V / R or 4.5 / 9.0 (1)
- 0.50 A (1)

ECF from (a) and (b)

(d) **[1]**

- 0.50 A and the current is the same at every point in a series circuit (1)

ECF from (c)

REJECT a different value or no reason

Worked solution

(a) The cells all point the same way, so their e.m.f.s add: 1.5 + 1.5 + 1.5 = 4.5 V.

(b) Resistances in series add: 2.0 + 7.0 = 9.0 Ω.

(c) R = V / I, so I = V / R = 4.5 V / 9.0 Ω = 0.50 A. Check: 0.50 A × 9.0 Ω = 4.5 V.

(d) There is only one path in a series circuit, so the current is the same everywhere: the 2.0 Ω resistor also has 0.50 A in it.

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17 · Part 3 Core Series and parallel circuits [6]

Mark scheme

(a) **[2]**

- each lamp can be switched on and off independently / separately (1)
- if one lamp fails (or is switched off), the others stay lit (1)

ALLOW each lamp has the full supply p.d. across it (so works at full brightness) (1)

- any two, max 2

(b) **[1]**

- larger (than 0.25 A) (1)

(c) **[1]**

- the combined resistance is less than the resistance of one lamp (1)

ALLOW smaller / half

(d) **[2]**

- (each lamp added in parallel) the combined resistance of the circuit decreases (1)
- the supply p.d. stays the same, so (by I = V / R) the current increases (1)

ALLOW each extra lamp adds its own branch current, so the total current increases (1)

Worked solution

(a) In parallel each lamp is on its own branch, connected directly across the supply. Each lamp can therefore have its own switch, a broken lamp does not turn the others off, and each lamp gets the full mains p.d. so it is at normal brightness.

(b) The supply current divides between the branches, so the current from the supply is larger than the current in any one branch.

(c) Two resistances in parallel have a combined resistance less than either one alone, because there are now two paths for the charge.

(d) Switching on more lamps adds more parallel paths, so the combined resistance falls. The mains p.d. does not change, so I = V / R gives a larger current from the supply.

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18 · Part 3 Core Series and parallel circuits [5]

Mark scheme

(a) **[2]**

- ammeter connected in series (with the lamp) (1)
- voltmeter connected in parallel with / across the resistor (1)

(b) **[1]**

- the current is the same at every point in a series circuit (1)

(c)(i) **[1]**

- increases (1)

(c)(ii) **[1]**

- decreases (1)

ALLOW the lamp becomes dimmer, only if "current decreases" is also stated

Worked solution

(a) An ammeter measures the current through it, so it must be in series, in the same loop as the lamp. A voltmeter measures the p.d. between two points, so it is connected in parallel, with one lead on each side of the resistor.

(b) All three readings are the same, which shows the current is the same at every point in a series circuit; it is not used up by the lamp.

(c)(i) Resistances in series add, so adding another resistor makes the combined resistance larger.

(c)(ii) The battery's e.m.f. is unchanged and the resistance is larger, so by I = V / R the current decreases, including the current in the lamp.

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19 · Part 3 Extended Series and parallel circuits [7]

Mark scheme

(a) **[2]**

- 1 / R = 1 / 20 + 1 / 30 (1)

ALLOW R = (20 × 30) / (20 + 30)

- 12 Ω (1)

(b) **[1]**

- 8.0 + 12 = 20 Ω (1)

ECF from (a)

(c) **[1]**

- I = 10 / 20 = 0.50 A (1)

ECF from (b)

(d) **[2]**

- p.d. across 8.0 Ω resistor = 0.50 × 8.0 = 4.0 V (1)
- p.d. across parallel arrangement = 10 − 4.0 = 6.0 V (1)

ALLOW 0.50 × 12 = 6.0 V

ECF from (c)

(e) **[1]**

- I = 6.0 / 30 = 0.20 A (1)

ECF from (d)

Worked solution

(a) For two resistors in parallel, 1 / R = 1 / 20 + 1 / 30 = 3 / 60 + 2 / 60 = 5 / 60, so R = 60 / 5 = 12 Ω. This is less than either 20 Ω or 30 Ω, as expected.

(b) The 8.0 Ω resistor is in series with the 12 Ω combination: 8.0 + 12 = 20 Ω.

(c) I = V / R = 10 V / 20 Ω = 0.50 A.

(d) p.d. across the 8.0 Ω resistor = IR = 0.50 A × 8.0 Ω = 4.0 V. The p.d.s round a series circuit add up to the supply p.d., so the parallel arrangement has 10 − 4.0 = 6.0 V across it (check: 0.50 A × 12 Ω = 6.0 V).

(e) Each branch of the parallel arrangement has the full 6.0 V across it: I = 6.0 V / 30 Ω = 0.20 A. Check: the 20 Ω branch carries 6.0 / 20 = 0.30 A, and 0.20 + 0.30 = 0.50 A, which equals the supply current.

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20 · Part 3 Extended Series and parallel circuits [4]

Mark scheme

(a) **[2]**

- current is a flow of charge / rate of flow of charge (1)
- charge is not created, destroyed or stored at the junction, so the charge flowing in each second equals the charge flowing out each second (1)

ALLOW charge is conserved (for the second mark)

(b) **[1]**

- 2.4 − (0.80 + 0.60) = 1.0 A (1)

(c) **[1]**

- Y, because it has the smallest current for the same p.d. (1)

REJECT Y with no reason

Worked solution

(a) Electric current is the rate of flow of charge. Charge cannot pile up at a junction, and it is not created or destroyed there, so every coulomb that arrives each second must leave each second. So total current in = total current out.

(b) Current out must total 2.4 A: current in Z = 2.4 − 0.80 − 0.60 = 1.0 A. Check: 0.80 + 0.60 + 1.0 = 2.4 A.

(c) R = V / I, and V is the same for all three, so the resistor with the smallest current has the largest resistance. Y carries the smallest current (0.60 A), so Y has the largest resistance.

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21 · Part 3 Extended Series and parallel circuits [6]

Mark scheme

(a)(i) **[1]**

- 24 + 24 = 48 Ω (1)

(a)(ii) **[1]**

- 12 Ω (1)

> **ALLOW** from 1 / R = 1 / 24 + 1 / 24 or (24 × 24) / (24 + 24)

(b) **[1]**

- I = 12 / 12 = 1.0 A (1)

> **ECF** from (a)(ii)

(c) **[2]**

- setting 1: I = 12 / 48 = 0.25 A, P = IV = 0.25 × 12 = 3.0 W (1)
- setting 2: P = IV = 1.0 × 12 = 12 W (1)

> **ECF** from (a) and (b)

(d) **[1]**

- 4 : 1 (1)

> **ALLOW** 4.0 : 1

> **ECF** from (c)

Worked solution

(a)(i) In series, resistances add: 24 + 24 = 48 Ω.

(a)(ii) In parallel, 1 / R = 1 / 24 + 1 / 24 = 2 / 24, so R = 12 Ω. Two equal resistors in parallel give half the resistance of one.

(b) I = V / R = 12 V / 12 Ω = 1.0 A.

(c) Setting 1: I = 12 V / 48 Ω = 0.25 A, so P = IV = 0.25 A × 12 V = 3.0 W. Setting 2: P = IV = 1.0 A × 12 V = 12 W. Check setting 2 another way: each element has the full 12 V across it and carries 12 / 24 = 0.50 A, so each gives 6.0 W, a total of 12 W.

(d) 12 W : 3.0 W = 4 : 1. The parallel setting gives four times the power, so it is the high setting.

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22 · Part 3 Extended Action and use of circuit components [6]

Mark scheme

(a) **[2]**

- the output p.d. increases (steadily) as S moves from X to Y (1)
- from 0 V (at X) to 12 V (at Y) (1)

(b) **[3]**

- resistance between S and Y = 600 − 150 = 450 Ω (1)
- V(XS) / V(SY) = 150 / 450 or output = 150 / 600 × 12 (1)
- 3.0 V (1)

ALLOW current = 12 / 600 = 0.020 A, output = 0.020 × 150 = 3.0 V (3)

(c) **[1]**

- any one sensible use, e.g. volume control / dimmer switch for a lamp / controlling the speed of a motor / giving an adjustable p.d. for an experiment (1)

Worked solution

(a) The track divides the 12 V supply between the part X to S and the part S to Y in the ratio of their resistances. With S at X there is no resistance between X and S, so the output is 0 V. As S moves towards Y, the resistance between X and S increases, so the output p.d. increases steadily until it is 12 V when S reaches Y.

(b) The two parts of the track act as two resistors in series: 150 Ω (X to S) and 450 Ω (S to Y). Using R1 / R2 = V1 / V2, V(XS) / V(SY) = 150 / 450 = 1 / 3, so the output is one quarter of the total: 12 V × 1 / 4 = 3.0 V. Check: the current in the track is 12 V / 600 Ω = 0.020 A, and 0.020 A × 150 Ω = 3.0 V.

(c) A variable potential divider gives an output p.d. that can be set anywhere between zero and the supply p.d., for example as a volume control or a lamp dimmer.

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23 · Part 3 Extended Action and use of circuit components [6]

Mark scheme

(a) **[2]**

- V(LDR) / V(resistor) = 27 / 3.0 or V(LDR) = 27 / 30 × 6.0 (1)
- 5.4 V (1)

ALLOW I = 6.0 / 30 000 = 0.00020 A, V = 0.00020 × 27 000 = 5.4 V

(b) **[2]**

- V(LDR) / V(resistor) = 1.0 / 3.0 or V(LDR) = 1.0 / 4.0 × 6.0 (1)
- 1.5 V (1)

(c) **[2]**

- the fixed resistor (1)
- the current is the same in both (series) and the fixed resistor has the larger resistance (3.0 kΩ compared with 1.0 kΩ) (1)

ALLOW fixed resistor has 4.5 V compared with 1.5 V across the LDR, with reason linked to resistance

Worked solution

The LDR and the fixed resistor form a potential divider, so the 6.0 V is shared in the ratio of their resistances (R1 / R2 = V1 / V2).

(a) Dark: total resistance = 27 + 3.0 = 30 kΩ. The LDR has 27 / 30 of the supply p.d.: 27 / 30 × 6.0 V = 5.4 V. Check: the fixed resistor has 0.60 V, and 5.4 / 0.60 = 9 = 27 / 3.0.

(b) Bright: total resistance = 1.0 + 3.0 = 4.0 kΩ. The LDR has 1.0 / 4.0 of the supply p.d.: 1.0 / 4.0 × 6.0 V = 1.5 V. The fixed resistor has 4.5 V.

(c) In series both components have the same current. For a constant current the p.d. is larger across the larger resistance, so in bright sunlight the 3.0 kΩ fixed resistor has the larger p.d. (4.5 V).

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24 · Part 3 Core Electrical safety [5]

Mark scheme

(a) **[3]**

- a large current flows from the live wire through the casing and the earth wire (to earth) (1)
- this (large) current makes the fuse melt / blow (1)
- this breaks the circuit / disconnects the live wire, so the casing is no longer live and the person does not get a shock (1)

ALLOW the earth wire is a low-resistance path, so current passes through it rather than through the person

(b) **[1]**

- the casing is made of an insulator / does not conduct, so it cannot become live (1)

(c) **[1]**

- the circuit and the cabling (of the appliance) (1)

ALLOW the cable / flex from overheating

ALLOW the appliance (from too large a current)

Worked solution

(a) The earth wire connects the metal casing to earth through a path of very low resistance. If the live wire touches the casing, a large current flows from the live wire through the casing and down the earth wire. This current is much larger than the fuse rating, so the fuse melts and breaks the circuit. The appliance is then disconnected from the live wire, so the casing cannot give anyone a shock.

(b) A double-insulated appliance has a non-conducting outer casing, so even if a live wire comes loose inside, the casing cannot become live. An earth wire is not needed.

(c) Without an earth wire, the fuse still protects the circuit and the cabling: if too large a current flows, the fuse melts before the cable overheats.

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25 · Part 3 Core Electrical safety [7]

Mark scheme

(a) **[2]**

- I = P / V or 2760 / 230 (1)
- 12 A (1)

ALLOW 12.0 A

(b) **[2]**

- total power = 2760 + 920 + 1150 = 4830 W or currents 12 A + 4.0 A + 5.0 A (1)
- 4830 / 230 = 21 A (1)

ECF from (a) for the current-adding method

(c) **[2]**

- the trip switch trips / opens / switches off and cuts off the current (1)
- because the current (21 A) is larger than its setting of 20 A, so the cables do not overheat / no fire risk (1)

ECF from (b)

(d) **[1]**

- it can be reset (by a switch) instead of being replaced (1)

ALLOW it acts faster than a fuse

Worked solution

(a) P = IV, so I = P / V = 2760 W / 230 V = 12 A.

(b) The total power is 2760 + 920 + 1150 = 4830 W, so the total current is 4830 W / 230 V = 21 A. Check another way: toaster 920 / 230 = 4.0 A, microwave 1150 / 230 = 5.0 A, and 12 + 4.0 + 5.0 = 21 A.

(c) 21 A is more than the 20 A setting, so the trip switch trips and disconnects the circuit. This stops a current that could overheat the cables and start a fire.

(d) After the fault is removed (here, by not using all three at once), a trip switch is simply reset, whereas a melted fuse must be replaced. Trip switches also act more quickly than fuses.

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26 · Part 3 Core Electrical safety [4]

Mark scheme

(a) **[1]**

- electric shock (from touching the exposed live wire) (1)

ALLOW fire from sparking / short circuit

(b) **[2]**

- the extension lead (or plug / socket) is overloaded, so the (total) current in it is very large (1)
- the cable / lead overheats, which can melt insulation or cause a fire (1)

(c) **[1]**

- live (line), neutral and earth (1)
- all three needed for the mark

Worked solution

(a) Damaged insulation can expose the metal of a live wire. Touching it gives an electric shock.

(b) All three appliances draw current through the one extension lead and the one wall socket, so the total current is large. A large current heats the cable, which can melt the insulation and cause a fire.

(c) A mains cable has a live (line) wire, a neutral wire and an earth wire.

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27 · Part 3 Core Electromagnetic induction [5]

Mark scheme

(a) **[3]**

- connect the coil to the ammeter to make a complete circuit (1)
- move the magnet into (or out of) the coil (1)
- the ammeter needle deflects (shows a reading) while the magnet is moving (1)

ALLOW deflection in the opposite direction when the magnet is moved out

(b) **[1]**

- the magnetic field linking the coil is not changing / no field lines are cut, so no e.m.f. is induced (1)

(c) **[1]**

- move the coil towards / away from (over) the magnet (1)

Worked solution

(a) Connect the two ends of the coil to the centre-zero ammeter. Push the magnet into the coil: the needle deflects, showing an induced current. Pull the magnet out: the needle deflects the other way. Moving the magnet faster gives a larger deflection.

(b) An e.m.f. is induced only when the magnetic field linking the coil changes. If the magnet is still, the field through the coil is constant, so there is no e.m.f. and no current.

(c) What matters is relative motion: keeping the magnet still and moving the coil towards or away from it also changes the field linking the coil, so an e.m.f. is induced.

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28 · Part 3 Core Electromagnetic induction [4]

Mark scheme

(a) **[1]**

- more turns gives a larger e.m.f. (1)

ALLOW doubling the number of turns doubles the e.m.f.

(b) **[1]**

- a higher speed gives a larger e.m.f. (1)

(c) **[1]**

- 2.2 V (1)

ALLOW "about double 1.10 V" or any value from 2.0 V to 2.4 V

(d) **[1]**

- the strength of the magnet (1)

REJECT number of turns or speed (already tested)

Worked solution

(a) Tests 1 and 2 have the same speed but the number of turns doubles from 300 to 600, and the e.m.f. doubles from 0.40 V to 0.80 V. More turns give a larger induced e.m.f.

(b) Tests 1 and 3 have the same number of turns, but the faster magnet gives 1.10 V instead of 0.40 V. A higher speed gives a larger e.m.f.

(c) Test 3 shows 1.10 V for 300 turns at high speed. Tests 1 and 2 show that doubling the turns doubles the e.m.f., so 600 turns at the same high speed should give about $2 \times 1.10 = 2.2$ V.

(d) The third factor is the strength of the magnetic field, for example using a stronger magnet.

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29 · Part 3 Extended Electromagnetic induction [5]

Mark scheme

(a) **[2]**

- a north pole (1)
- the induced e.m.f. / current opposes the change causing it, so end P repels the approaching north pole (1)

(b) **[1]**

- a south pole (1)

(c) **[2]**

- there is a (repulsive) force opposing the motion of the magnet, so work must be done against it (1)
- this work is transferred to electrical energy in the circuit (and then to thermal energy in the resistor); energy is conserved (1)

Worked solution

(a) The direction of an induced e.m.f. always opposes the change causing it. The change is a north pole coming closer, so the induced current makes end P a north pole, which pushes back on the approaching magnet.

(b) Now the change is the north pole moving away. To oppose this, end P becomes a south pole, which attracts the magnet and tries to hold it back.

(c) Because the coil always pushes against the motion, the person must exert a force over a distance, which is work done. Energy cannot be created: this work is the source of the electrical energy carried by the induced current, which is then transferred to the surroundings as thermal energy in the resistor. If there were no opposing force, electrical energy would appear from nothing.

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30 · Part 3 Extended Electromagnetic induction [6]

Mark scheme

(a) **[1]**

- towards the bottom of the page (1)

(b) **[1]**

- the same size of reading but in the opposite direction (1)

(c) **[2]**

- no e.m.f. is induced (1)
- the wire does not cut (move across) any magnetic field lines / it moves parallel to its own length (1)

(d) **[2]**

- move the wire faster (1)
- use a stronger magnetic field (1)

ALLOW a longer length of wire in the field (1)

- any two, max 2

Worked solution

(a) Use Fleming's right-hand rule. Point the thumb in the direction of motion (towards the left of the page) and the first finger along the field (into the page). The second finger points towards the bottom of the page, so the induced current flows down the wire.

(b) Reversing the direction of motion reverses the induced current. The speed is unchanged, so the size of the e.m.f. is unchanged: the ammeter shows the same size of reading in the opposite direction.

(c) An e.m.f. is induced only when the wire cuts across field lines. Moving the wire along its own length does not cut any field lines, so no e.m.f. is induced along the wire and the ammeter reads zero.

(d) The induced e.m.f. is larger when the wire cuts field lines faster: move the wire faster, or use a stronger magnetic field (more field lines cut each second).

1 · Part 4 Extended The a.c. generator [1]

Answer: B

Worked solution

In a rotating-coil generator the coil turns continuously. If its ends were fixed wires they would wind up and break. Each end of the coil is joined to its own ring; a carbon brush presses on each ring and carries the current to the circuit. Because each ring stays joined to the same end of the coil, the output stays alternating.

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2 · Part 4 Core Magnetic effect of a current [1]

Answer: D

Worked solution

If iron filings are sprinkled on the card and the card is tapped, they line up in rings around the wire. The rings are closer together near the wire, where the field is strongest.

3 · Part 4 Core Magnetic effect of a current [1]

Answer: A

Worked solution

Use the right-hand grip rule: curl the fingers of the right hand in the direction of the current; the thumb points to the north pole. With the current anticlockwise as seen from the left, the thumb points out of the left end, towards the observer, so the field lines leave the left end. Field lines leave a north pole, so the left end is north.

4 · Part 4 Extended Magnetic effect of a current [1]

Answer: C

Worked solution

The field lines around a straight wire are circles that are close together near the wire and spread further apart with distance. Closer field lines mean a stronger field, so the field strength falls as you move away from the wire.

5 · Part 4 Core Magnetic effect of a current [1]

Answer: A

Worked solution

The coil and core form an electromagnet. When the small current is switched on, the electromagnet pulls on a pivoted iron armature, which pushes two contacts together, so the large current can flow in the second circuit. When the small current stops, a spring opens the contacts again.

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6 · Part 4 Core Force on a current-carrying conductor [1]

Answer: B

Worked solution

Each reversal flips the direction of the force. One flip makes it downwards; a second flip makes it upwards again. So the wire is pushed upwards, just as before.

7 · Part 4 Extended Force on a current-carrying conductor [1]

Answer: D

Worked solution

Fleming's left-hand rule: First finger = Field, seCond finger = Current, thuMb = Motion (force). Point the first finger into the page and the second finger to the right; the thumb then points up the page. The force is always at right angles to both the current and the field.

8 · Part 4 Extended Force on a current-carrying conductor [1]

Answer: C

Worked solution

Step 1: the conventional current direction is opposite to the electron motion, so the current is towards the top of the page.

Step 2: Fleming's left-hand rule with the first finger into the page and the second finger up the page gives the thumb pointing to the left.

So the beam curves towards the left of the page.

9 · Part 4 Core The d.c. motor [1]

Answer: B

Worked solution

The turning effect on a coil increases with the number of turns, the current and the strength of the magnetic field. Changing the direction of the current only changes which way the coil turns.

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10 · Part 4 Core The transformer [1]

Answer: A

Worked solution

The two coils are insulated and are not electrically connected to each other. They are linked only by the soft-iron core, which is easily magnetised and demagnetised and carries the changing magnetic field from the primary coil to the secondary coil.

11 · Part 4 Core The transformer [1]

Answer: D

Worked solution

Vp / Vs = Np / Ns, so Vs = Vp × Ns / Np = 120 × 60 / 1200 = 6.0 V.

Check: the secondary has 20 times fewer turns than the primary, so the voltage is 20 times smaller: 120 / 20 = 6.0 V. This is a step-down transformer.

12 · Part 4 Core The transformer [1]

Answer: C

Worked solution

Vp / Vs = Np / Ns, so Ns = Np × Vs / Vp = 800 × 400 kV / 25 kV = 800 × 16 = 12 800 turns.

Check: a step-up transformer must have more turns on the secondary than the primary, and 12 800 is 16 times 800, matching the voltage increase of 16 times.

13 · Part 4 Core The transformer [1]

Answer: A

Worked solution

The power delivered depends on current × voltage. Raising the voltage lets the same power be sent with a much smaller current. The heating of the cables depends on the current, so a smaller current wastes less energy as heat.

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14 · Part 4 Extended The transformer [1]

Answer: C

Worked solution

Power out = IsVs = 4.0 × 11.5 = 46 W. For 100% efficiency the power in is also 46 W.

Ip = 46 / 230 = 0.20 A.

Check: the voltage is stepped down by a factor of 20, so the current is stepped up by the same factor going from primary to secondary: 0.20 A × 20 = 4.0 A.

15 · Part 4 Extended The transformer [1]

Answer: B

Worked solution

Power transmitted = IV, so with the same power, I is inversely proportional to V. Voltage × 5 means current ÷ 5.

Power lost in the cables = I^2R. With R unchanged, I ÷ 5 means I^2 ÷ 25, so the power lost becomes 25 times smaller.

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16 · Part 4 Extended The a.c. generator [5]

Mark scheme

(a) **[2]**

- (i) slip rings (1)
- (ii) brushes (1)

ALLOW carbon brushes

(b) **[2]**

- the coil rotates (continuously), so wires fixed to it would twist / tangle / break (1)
- the brushes slide on the rings, so the coil stays connected to the lamp / the circuit stays complete as the coil turns (1)

(c) **[1]**

- the coil does not move, so wires can be connected directly to it / there are no moving electrical connections (1)

Worked solution

(a) The rings are slip rings and the carbon blocks are brushes.

(b) The coil turns round and round. Wires joined directly to its ends would wind up and soon break. Each slip ring turns with the coil while its brush stays still and rubs against it, so there is always a path for the current between the coil and the lamp.

(c) In a rotating-magnet generator the coil is fixed, so its ends can be wired straight to the circuit. Only the magnet moves, and the magnet is not part of the circuit, so no slip rings or brushes are needed.

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17 · Part 4 Core Magnetic effect of a current [6]

Mark scheme

(a) **[2]**

- sprinkle iron filings (thinly) on the card around the wire (1)
- switch on the current and tap the card gently (so the filings line up with the field) (1)

(b) **[1]**

- circles centred on the wire (1)

ALLOW concentric circles / rings around the wire

IGNORE "closer together near the wire"

(c) **[2]**

- place a plotting compass on the card (near the wire) (1)
- the direction in which the north (-seeking) end of the needle points is the field direction; move the compass to different places around the wire / mark the direction at each place (1)

(d) **[1]**

- clockwise (1)

Worked solution

(a) Sprinkle a thin layer of iron filings on the card. Switch on the current and gently tap the card so the filings can move and line up along the field lines.

(b) The filings form circles around the wire, with the wire at the centre.

(c) Iron filings show the shape but not the direction. A small plotting compass placed on the card lines up with the field; its north end points along the field direction. Move it around the wire and mark the direction at several places.

(d) Use the right-hand grip rule: point the right thumb in the direction of the current (downwards, away from the student looking from above). The fingers curl clockwise as seen from above, so the field lines are clockwise.

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18 · Part 4 Extended Magnetic effect of a current [6]

Mark scheme

(a) **[1]**

- north pole (1)

(b) **[2]**

- (i) straight, parallel (and evenly spaced) lines along the length / axis of the solenoid (1)

ALLOW uniform field

- (ii) the field is stronger inside the solenoid (1)

(c) **[1]**

- the field becomes stronger (1)

ALLOW field lines closer together

IGNORE references to the shape of the field

(d) **[2]**

- the direction of the magnetic field is reversed (1)
- end X becomes a south pole (1)

ALLOW the poles swap / the compass needle reverses (for 1 mark only, if neither point above is stated)

Worked solution

(a) The north end of a compass points along the field. Outside a magnet, field lines point away from a north pole, so end X is a north pole.

(b) Inside a long solenoid the field lines are straight, parallel and evenly spaced, so the field is uniform. The field lines are much closer together inside than outside the side of the solenoid, so the field is stronger inside.

(c) A larger current gives a stronger magnetic field.

(d) Reversing the current reverses the direction of the field everywhere. The shape of the pattern stays the same, but the poles swap: X becomes a south pole.

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19 · Part 4 Core Magnetic effect of a current [5]

Mark scheme

(a) **[1]**

- the coil is a current-carrying conductor in a magnetic field (so a force acts on it) (1)

ALLOW the magnetic field of the current in the coil interacts with the field of the permanent magnet

(b) **[3]**

- the alternating current keeps changing direction (1)
- so the force on the coil keeps changing direction / the coil (and cone) moves backwards and forwards / vibrates (1)
- the vibrating cone pushes and pulls on the air, producing compressions and rarefactions / a sound wave (1)

(c) **[1]**

- the pitch of the sound becomes higher (1)

REJECT louder

Worked solution

(a) A wire carrying a current in a magnetic field feels a force. The coil carries a current and sits in the magnet's field, so there is a force on it.

(b) An alternating current reverses direction many times each second. Each time the current reverses, the force on the coil reverses, so the coil and the cone fixed to it move in and out, or vibrate. The vibrating cone squashes and stretches the air in front of it, sending out compressions and rarefactions: a sound wave.

(c) The cone vibrates at the same frequency as the current. A higher frequency gives a higher-pitched sound.

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20 · Part 4 Core Magnetic effect of a current [5]

Mark scheme

(a) **[1]**

- any one from:
- the large current would need thick (heavy, expensive) wires to and from the key switch (1)
- the key switch could overheat / spark / be damaged by the large current (1)
- it is safer for the driver / keeps the large-current circuit away from the driver (1)

(b) **[3]**

- there is a (small) current in the relay coil (1)
- the coil (and its soft-iron core) becomes an electromagnet / is magnetised (1)
- it attracts an iron armature, which closes the contacts, so there is a current in the starter motor circuit (1)

(c) **[1]**

- any suitable example in which a small current switches a larger or separate circuit, e.g. switching on a central heating system / an electric heater controlled by a thermostat circuit / a burglar alarm / car headlights / a high-power motor from a low-current control circuit (1)

REJECT a transformer

Worked solution

(a) The starter motor needs 150 A, which is 750 times more than the key circuit. Carrying that much current to the dashboard would need very thick cables, and the key switch would get hot and spark.

(b) Turning the key completes the small circuit, so there is a current in the relay coil. The coil and its soft-iron core become an electromagnet, which pulls an iron armature towards it. The armature pushes the relay contacts together, completing the starter motor circuit, so the large current flows and the motor turns.

(c) Relays are used whenever a small or low-power circuit must switch a large current, for example a central heating control, car headlights or an alarm system.

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21 · Part 4 Core Force on a current-carrying conductor [6]

Mark scheme

(a) **[1]**

- the rod moves / rolls along the rails (1)

(b) **[2]**

- there is a current in the rod and the rod is in a magnetic field (1)
- so a force acts on the rod (1)

> **ALLOW** the magnetic field of the current in the rod interacts with the field of the magnet (for the first mark)

(c) **[1]**

- the rod moves in the opposite direction (1)

(d) **[1]**

- the rod moves in the same direction as in (a) (1)

(e) **[1]**

- the rod does not move, because glass is an (electrical) insulator so there is no current (1)

> **REJECT** "the rod does not move" with no reason

Worked solution

(a) When the switch is closed, the rod rolls along the rails.

(b) The rails and rod form a complete circuit, so there is a current in the rod. A conductor carrying a current in a magnetic field has a force on it, and this force pushes the rod along the rails.

(c) Reversing the current reverses the force, so the rod rolls the other way.

(d) Reversing the field as well reverses the force a second time, so the rod moves in the same direction as in (a).

(e) Glass does not conduct electricity, so there is no current in the rod and no force acts on it. The rod stays still.

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22 · Part 4 Extended Force on a current-carrying conductor [6]

Mark scheme

(a) **[1]**

- out of the page (1)

(b) **[1]**

- towards the right of the page (1)

(c) **[4]**

- (i) towards the bottom of the page (1)
- (ii) towards the top of the page (1)
- (ii) β-particles are negatively charged (and α-particles are positively charged), so the force is in the opposite direction (1)

ALLOW (ii) β-particles moving to the right are equivalent to a conventional current to the left

- (iii) the beam is not deflected / goes straight through, because γ-rays have no charge (1)

REJECT (iii) "not deflected" with no reason

Worked solution

Use Fleming's left-hand rule: First finger = Field, seCond finger = Current (conventional, the direction a positive charge moves), thuMb = force (Motion).

(a) First finger to the right, second finger down the page: the thumb points out of the page.

(b) First finger into the page, second finger down the page: the thumb points to the right.

(c)(i) α-particles are positive, so their motion to the right is a conventional current to the right. First finger out of the page, second finger to the right: the thumb points towards the bottom of the page.

(ii) β-particles are electrons, which are negative. Their motion to the right is equivalent to a conventional current to the left, so the force is reversed: towards the top of the page.

(iii) γ-rays are electromagnetic waves with no charge, so a magnetic field exerts no force on them and they pass straight through.

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23 · Part 4 Core The d.c. motor [4]

Mark scheme

(a) **[3]**

- increase the number of turns on the coil (1)
- increase the current (1)
- use a stronger magnet / increase the strength of the magnetic field (1)

ALLOW increase the voltage / p.d. of the supply (as a way of increasing the current)

(b) **[1]**

- the coil turns in the opposite direction (1)

Worked solution

(a) The turning effect on a coil in a magnetic field is increased by more turns of wire on the coil, a larger current in the coil, and a stronger magnetic field.

(b) Reversing the current reverses the forces on the sides of the coil, so the coil turns the other way. The size of the turning effect does not change.

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24 · Part 4 Extended The d.c. motor [6]

Mark scheme

(a) **[2]**

- the two sides of the coil at right angles to the field carry current in opposite directions, so the forces on them are in opposite directions (one up, one down) (1)
- these forces produce a turning effect / moment about the axle (1)

(b) **[3]**

- the split-ring commutator reverses the direction of the current in the coil (1)
- every half turn / when the plane of the coil is vertical / at right angles to the field (1)
- so the forces on each side of the coil keep the turning effect in the same rotational direction (and the coil keeps turning the same way) (1)

ALLOW without the commutator, the coil would rock back and forth / stop at the vertical position

(c) **[1]**

- to make a (sliding) electrical contact between the fixed supply wires and the rotating commutator (1)

Worked solution

(a) Current flows along one long side of the coil and back along the other, so the currents in the two sides are in opposite directions. Each side is a current-carrying conductor in a magnetic field, so each has a force on it, and the forces are opposite: one side is pushed up and the other down. Together they turn the coil about the axle.

(b) After the coil has turned a quarter of a turn, its plane is vertical. If the current kept flowing the same way round the coil, the forces would now turn it back again, and the coil would rock and stop. The split ring turns with the coil; just as the coil passes the vertical position, the gaps in the ring pass the brushes and each half of the ring touches the other brush. This reverses the current in the coil every half turn. The side of the coil on the left always has its current in the same direction, so the turning effect is always in the same direction and the coil keeps rotating.

(c) The brushes are fixed. They press on the rotating commutator so that current can pass between the supply and the moving coil.

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25 · Part 4 Core The transformer [5]

Mark scheme

(a) **[3]**

- any three from:
- a primary coil (1)
- a secondary coil (1)
- coils made of insulated wire (1)
- (both coils) wound on a soft-iron core (1)
- the coils are not electrically connected to each other / are linked only by the core (1)

REJECT steel core

(b) **[2]**

- step-up (1)
- there are more turns on the secondary coil than on the primary coil (1)

ALLOW the output voltage is greater than the input voltage, because Ns > Np

REJECT a reason that does not refer to the numbers of turns

Worked solution

(a) A simple transformer has two separate coils of insulated wire, the primary and the secondary, wound on the same soft-iron core. The coils are not joined electrically; the core links them magnetically.

(b) The secondary has 2000 turns and the primary has 500 turns. The secondary has more turns, so the output voltage is larger than the input voltage: it is a step-up transformer (by a factor of 4).

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26 · Part 4 Core The transformer [6]

Mark scheme

(a) **[3]**

- Vp / Vs = Np / Ns (1)
- Ns = 1840 × 12 / 230 (1)
- 96 (turns) (1)

(b) **[2]**

- Vs = 230 × 24 / 1840 (1)
- 3.0 V (1)

(c) **[1]**

- step-down (1)

Worked solution

(a) Vp / Vs = Np / Ns, so Ns = Np × Vs / Vp = 1840 × 12 / 230 = 22 080 / 230 = 96 turns.

Check: 230 V / 1840 turns = 0.125 V per turn, and 12 / 0.125 = 96 turns.

(b) Vs = Vp × Ns / Np = 230 × 24 / 1840 = 3.0 V.

Check: 24 turns × 0.125 V per turn = 3.0 V.

(c) Both outputs are lower than the 230 V input (each secondary has fewer turns than the primary), so it is a step-down transformer.

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27 · Part 4 Core The transformer [5]

Mark scheme

(a) **[2]**

- (i) step-up (transformer) (1)
- (ii) step-down (transformer) (1)

(b) **[2]**

- any two from:
- smaller current in the cables (for the same power) (1)
- less energy / power is wasted as heat in the cables (1)
- thinner / lighter / cheaper cables can be used (1)

IGNORE "less resistance"

(c) **[1]**

- high voltages are dangerous / 275 kV would be unsafe in homes (1)

ALLOW appliances are designed to work at 230 V

Worked solution

(a) At the power station the voltage is raised from 20 kV to 275 kV, so a step-up transformer is used. Near the homes it is lowered to 230 V, so a step-down transformer is used.

(b) To transmit the same power at a higher voltage, the current needs to be smaller. A smaller current heats the cables less, so less energy is wasted. A smaller current also means thinner, lighter and cheaper cables can be used.

(c) A very high voltage would be extremely dangerous inside a home, so it is stepped down to a safer value that household appliances are designed for.

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28 · Part 4 Extended The transformer [5]

Mark scheme

(a) **[3]**

- the alternating current in the primary coil produces an alternating / changing magnetic field (1)
- the (soft-)iron core carries / links this changing magnetic field through the secondary coil (1)
- the changing magnetic field in the secondary coil induces an e.m.f. / voltage across it (1)

(b) **[2]**

- a steady (d.c.) current produces a constant / unchanging magnetic field (1)
- so the field through the secondary coil is not changing and no e.m.f. is induced (1)

ALLOW the brief flash happens because the field changes only while the current is switching on

Worked solution

(a) The alternating current in the primary coil keeps changing size and direction, so the magnetic field it produces keeps changing too. The soft-iron core guides this changing magnetic field through the secondary coil. A changing magnetic field in a coil induces an e.m.f. across it, so there is an alternating voltage across the secondary and the lamp lights.

(b) A battery gives a steady current, so once the current has risen to its final value the magnetic field in the core is constant. An e.m.f. is induced only while the magnetic field is changing. The field changes for a moment when the switch is closed, which causes the brief flash, but after that the field is steady and no e.m.f. is induced, so the lamp stays off.

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29 · Part 4 Extended The transformer [6]

Mark scheme

(a) **[2]**

- I = P / V or 60 / 24 (1)
- 2.5 A (1)

(b) **[3]**

- IpVp = IsVs (1)
- Ip = 2.5 × 24 / 240 (1)
- 0.25 A (1)

ALLOW Ip = 60 / 240 (input power = output power) for the first two marks

ECF from (a)

(c) **[1]**

- 10 : 1 (1)

ALLOW 10

Worked solution

(a) P = IV, so I = P / V = 60 / 24 = 2.5 A.

(b) For a 100% efficient transformer, IpVp = IsVs. Ip = IsVs / Vp = 2.5 × 24 / 240 = 0.25 A.

Check: power in = power out = 60 W, and 60 / 240 = 0.25 A.

(c) Np / Ns = Vp / Vs = 240 / 24 = 10, so the ratio is 10 : 1.

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30 · Part 4 Extended The transformer [8]

Mark scheme

(a) **[2]**

- I = P / V = 600 000 / 10 000 (1)
- 60 A (1)

(b) **[2]**

- $P = I^2R = 60^2 \times 5.0$ (1)
- 18 000 W (1)

ALLOW 18 kW / 1.8×10^4 W

ECF from (a)

(c) **[2]**

- I = 600 000 / 30 000 = 20 A (1)
- $P = 20^2 \times 5.0 = 2000$ W (1)

ALLOW 2.0 kW / 2.0×10^3 W

(d) **[2]**

- at the higher voltage the current is smaller (for the same power) (1)
- power wasted is proportional to I^2, so the loss is much smaller / is 9 times smaller (1)

ALLOW less energy wasted as heat, so more of the power reaches the town (for the second mark)

ECF from (b) and (c)

Worked solution

(a) Power = current × voltage, so I = P / V = 600 000 W / 10 000 V = 60 A.

(b) Power wasted in the cables = $I^2R = 60 \times 60 \times 5.0 = 18\,000$ W (18 kW). That is 3% of the power sent.

(c) At 30 kV: I = 600 000 / 30 000 = 20 A. Power wasted = $20 \times 20 \times 5.0 = 2000$ W (2.0 kW).

Check: the voltage is 3 times larger, so the current is 3 times smaller and the loss is $3^2 = 9$ times smaller: 18 000 / 9 = 2000 W.

(d) Sending the same power at a higher voltage needs a smaller current. Because the heating of the cables depends on the current squared, tripling the voltage cuts the wasted power by a factor of 9. Transmitting at high voltage therefore wastes much less energy in the cables.

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